7. **Using probes to identify specific DNA**
A radioactive probe (DNA labeled with a radioactive substance) is added to the DNA bands. The probe attaches by base pairing to those restriction fragments that are complementary to its sequence. The probes can also be prepared by using either 'fluorescent substance' or 'radioactive isotopes'.

8. **Hybridization with probe**
After the probe hybridizes and the excess probe washed off, a photographic film is placed on the membrane containing 'DNA hybrids'.

9. **Exposure on film to make a genetic/DNA Fingerprint**
The radioactive label exposes the film to form an image (image of bands) corresponding to specific DNA bands. The thick and thin dark bands form a pattern of **bars** which constitutes a genetic fingerprint.

Application of DNA finger printing

- **Forensic analysis** - It can be used in the identification of a person involved in criminal activities, for settling paternity or maternity disputes, and in determining relationships for immigration purposes.
- **Pedigree analysis** – inheritance pattern of genes through generations and for detecting inherited diseases.
- **Conservation of wild life** – protection of endangered species. By maintaining DNA records for identification of tissues of the dead endangered organisms.
- **Anthropological studies**–It is useful in determining the origin and migration of human populations and genetic diversities.

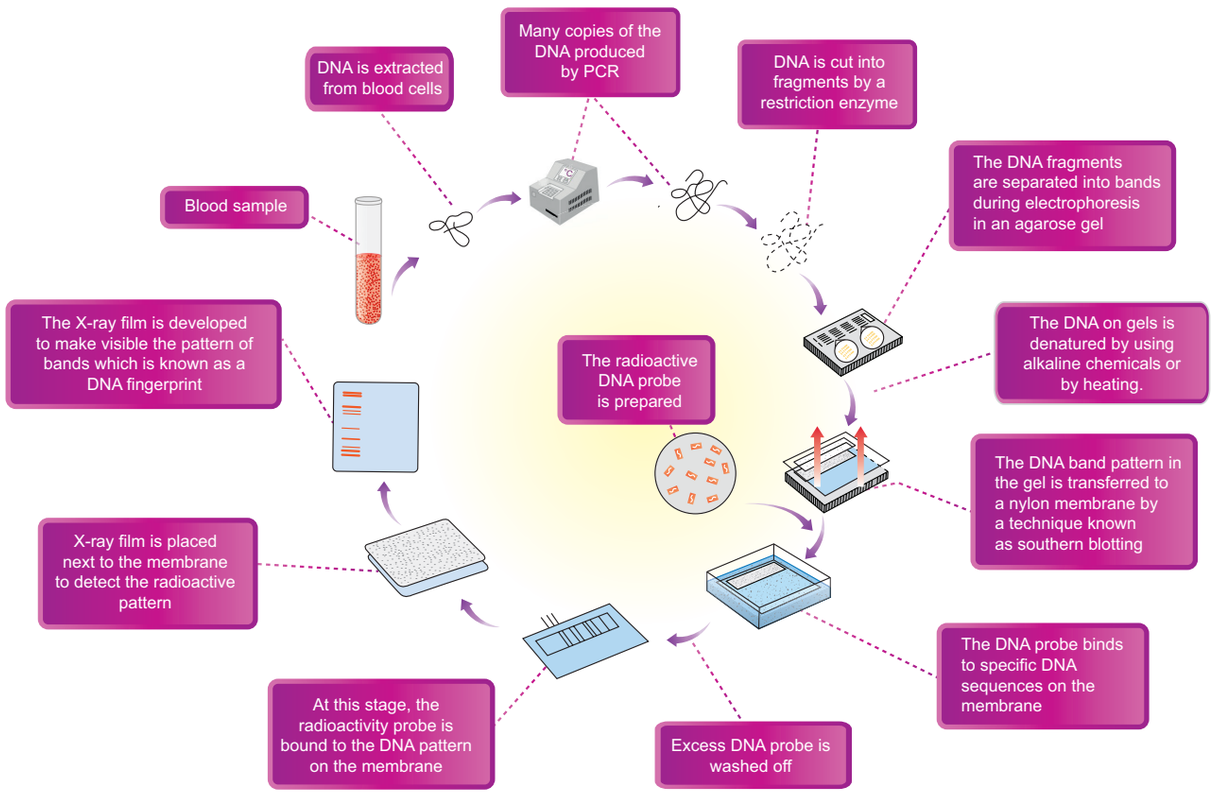


Fig. 5.16 The Steps involved in DNA Fingerprinting technique

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